GREENS TECHNOLOGYS

ONE COURSE. AN END-TO-END DATA STACK.

AWS DATA ENGINEERING

Build pipelines. Turn data into opportunity.



PLACEMENTS GUARANTEED

SQL | Python | AWS | Snowflake | dbt | PySpark | Power BI

AWS Data Engineering Master Course

126 live sessions | 189 hours | 29 modules

Since 2008 - 5,00,000+ careers launched

From foundations to data pipelines

Learn the data stack step by step, from SQL and Python to cloud pipelines, a warehouse and business dashboards.

SQL + Python | AWS + PySpark | Snowflake + dbt | Power BI

Build the foundations. Then connect the whole picture.

Extract data, transform it, load a warehouse and turn it into reports. Develop a project you can explain in interviews.

Who this course is for

Freshers | IT and non-IT graduates

Graduates from IT, engineering, arts, science, commerce and other degrees who want to start learning data engineering.

Non-IT employees | Moving into data

Working professionals changing careers who want to learn SQL, Python and cloud data skills from the foundations.

IT employees | Moving into data engineering

Developers, testers, support staff, database teams and cloud professionals who want to build data pipelines.

Data and analytics employees | Broadening their skills

SQL, BI, reporting and analytics professionals who want to add AWS, PySpark, Snowflake and dbt to their toolkit.

One course. A clear sequence.

Build confidence in the foundations, work through the cloud data stack and finish with an end-to-end capstone.

189 hours | 29 modules

126 live sessions | 4 capstone projects

01	Foundations (Modules 01-07)	34 sessions
02	AWS Core, Security & Storage (08-11)	12 sessions
03	Processing & Cataloguing (12-15)	14 sessions
04	Streaming & Orchestration (16-18)	13 sessions
05	Warehousing & Analytics (19-22)	18 sessions
06	Governance, Monitoring & DataOps (23-26)	11 sessions
07	BI & GenAI (27-29)	10 sessions
08	Capstone projects	8 sessions
09	Certification prep & mock exams	4 sessions
10	Career readiness	2 sessions

1

PHASE 1 OF 7 | MODULES 01-07 | 34 SESSIONS

Foundations

Pipelines, Python, SQL, PySpark, performance, partitioning and ETL/ELT

01

MODULE 01 · GREENS CORE

Data Pipeline Fundamentals

3 SESSIONS
40 TOPICS

DEA-C01
ALIGNED

The vocabulary and architecture every AWS data engineer uses, mapped to real AWS services. LAB: Whiteboard an AWS retail pipeline: sources, raw-silver-gold zones on S3, catalog, warehouse and dashboard.

The Data Engineer Role 1

- Role of a data engineer
- Data lifecycle
- Volume, velocity and variety
- Distributed computing basics
- Data structures: graphs and trees

Medallion on AWS 2

- Data catalog: Glue Data Catalog, SageMaker Catalog
- Data transformation and the silver layer
- Aggregation and the gold layer
- Data warehouse and reporting
- Medallion architecture

Streaming Overview 3

- Apache Kafka
- Kinesis Data Streams
- Amazon Data Firehose
- Amazon MSK
- Managed Service for Apache Flink
- Lambda and Kappa architectures

Data Systems 4

- OLTP vs OLAP
- Row vs columnar storage
- Structured, semi-structured, unstructured data
- Data warehouse vs data lake vs lakehouse
- Schema-on-read vs schema-on-write

Operating Pipelines 5

- Data quality
- Monitoring
- Orchestration
- Data lineage
- Data governance

Processing Patterns 6

- ETL vs ELT
- Batch processing
- Full load and incremental processing
- Change data capture (CDC)
- Batch vs streaming
- Stateful vs stateless processing

GenAI for Data Engineers 7

- Vectors and embeddings
- Amazon Bedrock
- LLMs in pipelines

Sources & Ingestion 8

- Source systems: databases, DynamoDB, OpenSearch
- APIs and files
- Data ingestion and the raw layer
- Data APIs for other systems
- Replayability of ingestion

MODULE 02 · GREENS CORE

02 Python

8 SESSIONS
52 TOPICS

DEA-C01
ALIGNED

Full Python for data engineering, from first variable to tested, AWS-connected scripts. LAB: Pull a paginated REST API, clean it with pandas, write Parquet with pyarrow and upload to S3 with Boto3, with logs and tests.

Python Basics 1

- Variables and data types
- List, tuple, set, dictionary
- Operators
- Input and output
- Type conversion

Strings 2

- f-strings
- Slicing
- String methods
- Regular expressions

Comprehensions & Modules 3

- List comprehension
- Dictionary comprehension
- Modules and packages
- Imports

Functions 4

- Functions and return values
- Lambda functions
- *args and **kwargs
- Variable scope
- Type hints

Errors, Files & Logging 5

- Exception handling
- File handling: JSON
- File handling: CSV
- Logging

Control Flow 6

- Conditions: if, elif, else
- for and while loops
- break, continue, pass
- match-case

MODULE 03 · GREENS CORE

03 SQL

8 SESSIONS
52 TOPICS
DEA-C01 ALIGNED

Every SQL command category through warehouse-grade patterns, practised on a banking dataset. LAB: Build an SCD Type 2 customer dimension with MERGE, then write window-function reports for running balances and top-N per region.

SQL Basics 1

- DDL, DML, DQL
- DCL and TCL
- Constraints
- Normalisation

Subqueries & CTEs 2

- Subqueries
- Correlated subqueries
- CTEs
- Recursive CTEs

Data Modelling 3

- SCD Type 1
- SCD Type 2
- SCD Type 3
- Star and snowflake schemas
- Facts and dimensions

Querying 4

- SELECT and WHERE
- DISTINCT and LIMIT
- LIKE, IN, BETWEEN
- Aliases
- ORDER BY
- CASE

Window Functions 5

- ROW_NUMBER, RANK, DENSE_RANK
- LEAD and LAG
- FIRST_VALUE, LAST_VALUE
- NTILE
- Window frames: ROWS BETWEEN
- Running totals

Functions 6

- NULL handling: COALESCE, NULLIF
- CAST
- String functions
- Date functions
- PIVOT and UNPIVOT
- JSON in SQL

Aggregation 7

- GROUP BY and HAVING
- Aggregate functions
- Grouping, rolling averages, pivoting

Database Objects 8

- Views
- Materialized views
- Stored procedures
- Temporary tables

Performance & Transactions 9

- Indexes
- EXPLAIN plans
- Query optimization
- Transactions
- Isolation levels
- Locks and blocking

Joins & Sets
10

- Inner, left, right, full joins
- Cross and self joins
- UNION
- INTERSECT and EXCEPT

Loading Patterns
11

- MERGE
- Upsert
- Deduplication
- Incremental SQL

MODULE 04 · GREENS CORE

04 PySpark

6 SESSIONS
36 TOPICS

DEA-C01
ALIGNED

Distributed processing with the DataFrame API, ready for Glue, EMR and Databricks. LAB: Ingest nested JSON with an explicit schema, explode and clean it, mask card numbers and write partitioned Parquet to S3.

Spark Basics & Architecture
1

- Driver and executors
- SparkSession
- Jobs, stages, tasks and the DAG
- Cluster managers

Cleaning & Types
2

- Null handling
- Data type conversion
- when / otherwise

Reading & Writing
3

- CSV, JSON, Parquet
- Explicit schemas (StructType)
- Read/write options
- Write modes
- Spark SQL and temp views

Advanced Functions
4

- Window functions
- UDFs and pandas UDFs
- map
- String and date functions
- pivot

Core Concepts
5

- DataFrame and RDD
- Transformations vs actions
- Lazy evaluation

Beyond the Basics
6

- Broadcast variables
- Accumulators
- Structured Streaming
- Testing Spark code

DataFrame Operations
7

- select, filter, withColumn
- groupBy and agg
- orderBy / sort
- Joins and union
- Deduplication

JSON Processing
8

- explode
- Array functions
- Struct functions

Spark Platforms
9

- Spark on Glue
- Spark on Amazon EMR
- Databricks: Delta Lake
- Databricks notebooks and workflows

MODULE 05 · GREENS CORE

05 PySpark Performance

4 SESSIONS
24 TOPICS
DEA-C01 ALIGNED

Diagnosing slow Spark jobs and fixing them: the skill interviewers test hardest. LAB: Diagnose a slow skewed join in the Spark UI and cut runtime with broadcast, salting, AQE and right-sized files.

Partitions & Shuffle 1

- repartition and coalesce
- Shuffle
- Narrow vs wide transformations

Pruning 2

- Predicate pushdown
- Partition pruning
- Column pruning
- Dynamic partition pruning

Optimizer 3

- Adaptive Query Execution (AQE)
- Catalyst optimizer
- Tungsten
- Explain plans
- Reading the Spark UI

Joins 4

- Broadcast joins
- Skew join hints
- Bucketing

Data Skew & Files 5

- Data skew and salting
- Small file problem
- File-size tuning

Memory Tuning 6

- Executor cores and memory
- Spill
- Out-of-memory errors
- Memory optimization

Caching & Serialization 7

- Cache and persistence
- Kryo serialization

MODULE 06 · GREENS CORE

06 Partitioning

2 SESSIONS
19 TOPICS
DEA-C01 ALIGNED

Laying out data on S3 so Athena, Glue, Redshift Spectrum and Spark read only what they need. LAB: Partition a year of transactions three ways, compare Athena scan size and cost, and add partition projection.

Concepts 1

- What partitioning is
- Why partition
- Partition strategy
- Partition vs bucketing

Catalog & Discovery 2

- Partition pruning
- Partition discovery
- MSCK REPAIR TABLE
- ALTER TABLE ADD PARTITION
- Athena partition projection
- Glue partition indexes

Engines & Formats 3

- Spark partitioning
- Compaction and partition limits
- Iceberg hidden partitioning
- Partition evolution
- Redshift distribution and sort keys

S3 Partitioning 4

- Hive-style partitions
- Date partitioning: year / month / day
- Static vs dynamic partitioning
- High-cardinality and over-partitioning

MODULE 07 · GREENS CORE

07 ETL / ELT

3 SESSIONS
28 TOPICS

DEA-C01
ALIGNED

Production load patterns and the AWS services that move data in. LAB: Run a DMS full-load-plus-CDC task from Aurora to S3, with idempotent merges and a quarantine path for bad rows.

Patterns 1

- ETL vs ELT: extract, transform, load
- Batch ETL
- Full load vs incremental load
- CDC
- Watermarks and high-water marks

Data Correctness 2

- Data validation
- Reconciliation
- Audit columns
- Schema evolution
- Late-arriving data
- SCD loads
- Lineage

AWS Ingestion Services 3

- AWS DMS and DMS Schema Conversion
- Amazon AppFlow
- AWS DataSync
- AWS Transfer Family
- Snow Family overview
- Zero-ETL integrations

Reliability 4

- Idempotency
- Retry and reprocessing
- Backfills
- Error handling and quarantine
- Dead-letter handling

Streaming & Ops 5

- Streaming ETL
- Logging
- Alerting
- IP allowlists for sources
- Throttling and rate limits

2

PHASE 2 OF 7 | MODULES 08-11 | 12 SESSIONS

AWS Core, Security & Storage

IAM, networking, encryption, Amazon S3 and purpose-built databases

08

MODULE 08 · GREENS CORE

AWS IAM

3 SESSIONS

30 TOPICS

DEA-C01
ALIGNED

Identity and access for data platforms: least privilege, roles and cross-account access. LAB: Build least-privilege roles for Glue, Lambda and Redshift, assume a cross-account role and fix an AccessDenied with Access Analyzer.

Identities 1

- IAM users, groups and roles
- Root user protection and MFA
- IAM Identity Center (SSO)
- Access keys best practice and rotation

Roles & Credentials 2

- Service roles
- Role assumption
- Cross-account access
- STS temporary credentials
- Instance profiles

Data Service Access 3

- S3 IAM access
- Glue IAM access
- Lambda IAM access
- Redshift IAM access

Policies 4

- Trust policy vs permissions policy
- Policy JSON: Effect, Action, Resource, Condition
- Managed vs inline policies
- Resource-based policies
- Least privilege
- Custom policies

Troubleshooting 5

- AccessDenied troubleshooting
- IAM Access Analyzer
- Policy Simulator

Advanced Access 6

- ABAC with tags
- Permission boundaries
- SCPs and AWS Organizations
- Managed vs unmanaged services

Data Security Services 7

- KMS
- Secrets Manager
- Macie
- VPC endpoints

MODULE 09 ·

09 Networking, Encryption & Secrets

2 SESSIONS
19 TOPICS
DEA-C01 ALIGNED

The network and key-management layer that keeps data pipelines private and compliant. LAB: Put a Glue job and Lambda in private subnets with S3 and Secrets Manager endpoints, encrypted with a customer-managed KMS key.

VPC for Data Engineers 1

- VPC, subnets, route tables
- Security groups and NACLs
- NAT gateway
- Gateway vs interface endpoints
- AWS PrivateLink
- Glue and Lambda in a VPC

Encryption with KMS 2

- Customer-managed vs AWS-managed keys
- Key policies and grants
- Envelope encryption
- Cross-account encryption
- Encryption in transit (TLS)

Compliance 3

- Data sovereignty and Region restrictions
- Blocking replication to disallowed Regions
- Shield and WAF overview
- AWS Backup

Secrets 4

- Secrets Manager rotation
- Systems Manager Parameter Store
- Choosing between them
- Retrieving secrets in Glue and Lambda

MODULE 10 · GREENS CORE

10 Amazon S3

4 SESSIONS
33 TOPICS
DEA-C01 ALIGNED

The data lake foundation: storage, security, formats and cost on Amazon S3. LAB: Build a raw/silver/gold lake with SSE-KMS, lifecycle rules, an event notification to Lambda and an S3 Tables (Iceberg) table.

S3 Basics 1

- Buckets, objects, prefixes
- Multipart upload
- Consistency model

Security 2

- SSE-S3, SSE-KMS, SSE-C
- Block Public Access
- Object Ownership (ACLs disabled by default)
- ACLs vs bucket policies
- Bucket policy vs IAM policy

Formats & Performance 3

- CSV, JSON, Parquet, ORC, Avro
- Compression
- Small file problem
- S3 performance

Storage Classes & Lifecycle 4

- Storage classes
- Glacier tiers
- Intelligent-Tiering
- Lifecycle policies
- Expiring data by age

Access & Events 5

- Presigned URLs
- Access Points
- VPC endpoints
- Event notifications to Lambda, SQS, SNS, EventBridge

Data Lake on S3 6

- Raw, silver and gold layers
- Apache Iceberg
- Delta Lake
- S3 Tables
- S3 partitioning

Data Protection 7

- Versioning
- Cross- and same-Region replication
- Object Lock
- Deleting data for legal requirements

Cost & Visibility 8

- S3 cost optimization
- S3 Inventory
- Storage Lens

11 MODULE 11 · Purpose-Built Databases for Data Engineers

3 SESSIONS
20 TOPICS

DEA-C01 ALIGNED

Choosing and loading the right store for each access pattern, a core exam domain. LAB: Model an orders table in DynamoDB with TTL and Streams, and compare it with the same workload on Aurora.

Relational 1

- Amazon RDS
- Amazon Aurora
- Read replicas and Multi-AZ
- pgvector and HNSW on Aurora PostgreSQL
- JDBC and ODBC connections

Amazon DynamoDB 2

- Partition and sort keys
- GSIs and LSIs
- Capacity modes and throttling
- TTL
- DynamoDB Streams
- Export to S3

Other Stores 3

- DocumentDB
- Neptune (graphs)
- Keyspaces
- MemoryDB for key/value
- OpenSearch Service

Choosing a Store 4

- Access patterns
- Cost vs performance
- Locks and concurrency
- Provisioned vs serverless trade-offs

3

PHASE 3 OF 7 | MODULES 12-15 | 14 SESSIONS

Processing & Cataloguing

AWS Glue, crawlers, Amazon EMR and AWS Lambda

12

MODULE 12 · GREENS CORE

AWS Glue

6 SESSIONS
34 TOPICS

DEA-C01
ALIGNED

Serverless ETL on AWS: jobs, catalog, quality and cost control. LAB: Build an incremental Glue job with bookmarks, DQDL rules and an Iceberg MERGE into the silver layer.

Glue Overview 1

- Glue overview and architecture
- Glue Data Catalog
- Databases and tables
- Glue connections

DynamicFrames 2

- DynamicFrame vs DataFrame
- ApplyMapping
- ResolveChoice
- Relationalize

Scaling & Cost 3

- Worker types
- Auto scaling
- Flex execution
- Glue cost optimisation

Lake Formation Basics 4

- Data lake governance
- Lake Formation permissions

Incremental Processing 5

- Job parameters
- Job bookmarks
- Incremental processing
- Pushdown predicates

Quality & Monitoring 6

- Glue monitoring
- Error handling
- Glue Data Quality (DQDL)
- Glue DataBrew
- Spark UI for Glue

Glue Jobs 7

- Spark ETL jobs
- Python shell jobs
- Ray jobs (closed to new customers Apr 2026)
- Streaming ETL

Orchestration 8

- Glue triggers
- Glue workflows

Open Table Formats 9

- Glue with Apache Iceberg
- Iceberg MERGE and compaction

Development 10

- Glue Studio visual ETL
- Interactive sessions
- Notebooks

MODULE 13 · GREENS CORE

13 Glue Crawler

2 SESSIONS
20 TOPICS

DEA-C01 ALIGNED

Automatic schema and partition discovery that keeps the Data Catalog current. LAB: Crawl a partitioned S3 dataset and a JDBC source, add a custom classifier and switch to event-based crawls.

Crawler Basics 1

- Schema discovery
- Metadata discovery
- Writing metadata to the Data Catalog
- Table grouping

Partitions 2

- Partition discovery
- Partition indexes

Configuration 3

- Built-in classifiers
- Custom classifiers
- Exclusion patterns
- Schema evolution
- Schema change policies

Scheduling 4

- Crawler schedules
- Incremental crawls
- S3 event-based crawls

Sources 5

- S3 crawlers
- Database (JDBC) crawlers
- Crawling Iceberg tables
- Crawling Delta tables

Operations 6

- Crawler troubleshooting
- Crawler cost control

MODULE 14 ·

14 Amazon EMR

3 SESSIONS
16 TOPICS

DEA-C01 ALIGNED

Managed big-data clusters for Spark workloads that outgrow Glue. LAB: Run the same PySpark job on EMR Serverless and on an EMR cluster with Spot, and compare cost and runtime.

EMR Options 1

- EMR on EC2
- EMR Serverless
- EMR on EKS
- Choosing EMR vs Glue vs Lambda

Running Jobs 2

- Steps and spark-submit
- EMR Studio notebooks
- Hive and Presto/Trino overview
- Glue Data Catalog as metastore

Clusters 3

- Primary, core and task nodes
- Instance fleets and Spot
- Managed scaling
- Bootstrap actions

Operations 4

- Logging to S3
- Troubleshooting failed steps
- Cost optimisation
- Containers: ECR, ECS and EKS basics

MODULE 15 · GREENS CORE

15 AWS Lambda

3 SESSIONS
27 TOPICS

DEA-C01 ALIGNED

Event-driven serverless processing for light transformations and glue code. LAB: Deploy with SAM a Lambda that validates S3 uploads, routes failures to a DLQ and triggers a Glue job.

Lambda Basics 1

- Functions, handlers, events
- Runtimes
- Environment variables
- IAM role
- Layers

Concurrency 2

- Cold starts
- Reserved concurrency
- Provisioned concurrency

Reliability 3

- Error handling and retry
- Dead-letter queues and destinations
- Idempotency

Triggers & Patterns 4

- S3 to Lambda triggers
- Lambda to S3 and Lambda to Glue
- Kinesis, SQS, EventBridge, DynamoDB Streams triggers
- Fan-in and fan-out

Packaging 5

- Packaging dependencies
- Container images
- Lambda in a VPC
- EFS mounts
- AWS SAM

Limits 6

- Timeout
- Memory and ephemeral storage
- 15-minute limit and other quotas

Invocation 7

- Sync vs async invocation
- Versions and aliases

Operations 8

- Powertools logging
- Lambda with Step Functions

PHASE 4 OF 7 | MODULES 16-18 | 13 SESSIONS

4 Streaming & Orchestration

Kinesis, MSK, Flink, Airflow on MWAA, Step Functions and EventBridge

MODULE 16 ·

16 Streaming on AWS: Kinesis, Firehose, MSK & Flink

5 SESSIONS
25 TOPICS
DEA-C01 ALIGNED

Real-time ingestion and processing, the largest gap in most AWS syllabi. LAB: Stream clickstream events to Kinesis, aggregate in Flink windows and land them in S3 (Iceberg) via Firehose.

Kinesis Data Streams 1

- Shards and partition keys
- On-demand vs provisioned
- Retention and replay
- Enhanced fan-out
- Throttling and hot shards
- KCL and producers

Amazon Data Firehose 2

- Delivery to S3, Redshift, OpenSearch
- Buffering
- Lambda transforms
- Format conversion to Parquet
- Dynamic partitioning
- Delivery to Iceberg tables

Managed Service for Apache Flink 3

- Flink concepts
- Windows: tumbling, sliding, session
- Stateful processing
- Exactly-once semantics

Streaming Patterns 4

- Calling Lambda from Kinesis
- DynamoDB Streams
- Redshift streaming ingestion
- Spark Structured Streaming on Glue
- Lambda vs Kappa in practice

Amazon MSK 5

- Kafka topics and partitions
- MSK Provisioned vs Serverless
- MSK Connect
- Consumer groups

MODULE 17 · GREENS CORE

17 Apache Airflow

5 SESSIONS
41 TOPICS
DEA-C01 ALIGNED

Workflow orchestration with Airflow 3 on Amazon MWAA. LAB: Orchestrate S3 > Glue > Redshift on MWAA with asset-driven scheduling, retries, branching and alerting.

Architecture 1

- Scheduler, workers, executor
- API server and metadata DB
- What changed in Airflow 3

Data Passing 2

- XCom
- Variables
- Connections

Advanced 3

- Jinja templating and macros
- Dynamic task mapping
- Deferrable operators
- Monitoring
- Testing DAGs

DAGs 4

- DAGs, DAG runs and tasks
- DAG structure and default_args
- TaskFlow API and Task SDK
- DAG versioning

Control Flow 5

- Sensors
- Branching
- Task groups
- Trigger rules
- Pools and priority

Amazon MWAA 6

- Managed Airflow on AWS
- Deploying DAGs to S3/MWAA
- Environment sizing

Operators 7

- PythonOperator
- BashOperator
- AWS and Glue operators
- Hooks

Reliability 8

- Retries
- Failure handling
- Backfill and catchup
- Deadline alerts (replace Airflow 2 SLAs)

Alternatives 9

- AWS Step Functions
- Amazon EventBridge

DataOps 10

- CI/CD with Jenkins and GitHub Actions
- Infrastructure as Code with Terraform
- Docker basics
- Kubernetes basics

Scheduling 11

- Dependencies
- Scheduling and cron
- Data-aware scheduling: assets (formerly datasets)
- Event-driven scheduling with Asset Watchers

MODULE 18 ·

18 Step Functions, EventBridge, SNS & SQS

3 SESSIONS
16 TOPICS

DEA-C01
ALIGNED

Serverless orchestration and messaging, tested heavily in DEA-C01. LAB: Build a Step Functions workflow triggered by EventBridge that runs Glue and Athena, with retries and SNS alerts.

AWS Step Functions 1

- State machines and ASL
- Standard vs Express workflows
- Task, Choice, Map, Parallel states
- Retry and Catch
- Service integrations: Glue, Lambda, Athena

Amazon EventBridge 2

- Event buses and rules
- Event patterns
- EventBridge Scheduler
- Pipes

SNS & SQS 3

- SNS topics and subscriptions
- SQS standard vs FIFO
- Visibility timeout and DLQs
- Alerts with SNS and SQS

Patterns 4

- Serverless workflows
- Choosing Step Functions vs MWAA vs Glue workflows
- Resilient, fault-tolerant pipelines

5

PHASE 5 OF 7 | MODULES 19-22 | 18 SESSIONS

Warehousing & Analytics

Amazon Redshift, Amazon Athena, Snowflake and dbt

19

MODULE 19 · GREENS CORE

Amazon Redshift

6 SESSIONS
40 TOPICS

DEA-C01
ALIGNED

The AWS cloud data warehouse, from architecture to tuning and security. LAB: Load a star schema with COPY, tune dist and sort keys, add RLS and masking, and share it to another namespace.

Architecture 1

- Clusters
- Leader and compute nodes
- Slices
- RA3 and managed storage

Table Design 2

- Distribution styles
- Distribution keys
- Sort keys
- Compression encoding

Integration 3

- Redshift Spectrum
- Federated queries
- Zero-ETL
- Streaming ingestion
- Data API

Redshift Serverless 4

- Workgroups
- Namespaces
- RPUs
- Provisioned vs serverless

Maintenance 5

- VACUUM
- ANALYZE
- Locks and concurrency

Sharing & Security 6

- Data sharing
- RBAC
- Column-level security
- Row-level security
- Dynamic data masking

Query Optimization 7

- Concurrency scaling
- WLM queues
- Short query acceleration
- EXPLAIN
- System tables

Objects 8

- Databases, schemas, tables
- Materialized views
- SUPER type
- Stored procedures

Operations 9

- Snapshots
- Resize

Loading & Unloading 10

- COPY from S3 with an IAM role
- Manifest files
- Auto-copy
- UNLOAD

MODULE 20 · GREENS CORE

20 Amazon Athena

3 SESSIONS
25 TOPICS

DEA-C01
ALIGNED

Serverless SQL on the lake, and how to keep it fast and cheap. LAB: Query the lake with partition projection, convert JSON to Parquet with CTAS and MERGE into an Iceberg table.

Athena Overview 1

- Serverless SQL on S3
- Glue Data Catalog integration
- External tables
- Athena vs Redshift

Writing Data 2

- CTAS
- INSERT INTO
- UNLOAD
- Views

Iceberg in Athena 3

- MERGE
- Time travel
- OPTIMIZE and VACUUM

Advanced 4

- Federated queries
- Lake Formation permissions
- JSON SerDes
- Athena Spark notebooks
- Analysing logs with Athena

Performance 5

- Partitioning
- Partition projection
- Parquet and columnar formats
- Query optimization
- Cost optimization

Administration 6

- Workgroups
- Query result location
- Encryption
- Query result reuse

MODULE 21 · GREENS CORE

21 Snowflake

5 SESSIONS
30 TOPICS

SnowPro Core
ALIGNED

Snowflake running on AWS, loaded straight from S3. LAB: Auto-ingest S3 files with Snowpipe, build streams and tasks for incremental MERGE and secure them with masking.

Architecture 1

- Databases and schemas
- Virtual warehouses
- Editions and credits

Loading from S3 2

- Stages and external stages
- S3 storage integration with an IAM role
- COPY INTO
- File formats
- Unloading to S3
- Snowpipe through SQS

Incremental Loading 3

- Streams and tasks
- MERGE
- Dynamic tables

Warehouses 4

- Warehouse sizing
- Multi-cluster warehouses
- Auto-suspend
- Resource monitors

Semi-Structured & Code 5

- VARIANT
- Snowpark
- Iceberg tables

Storage 6

- Micro-partitioning
- Clustering
- Time Travel
- Fail-safe
- Zero-copy cloning

Performance & Security 7

- Query profile
- Result cache
- RBAC
- Masking policies
- Row access policies
- Data sharing

MODULE 22 · GREENS CORE

22 dbt (Data Build Tool)

4 SESSIONS
25 TOPICS

dbt
ALIGNED

The transformation layer of the modern data stack, on Redshift and Snowflake. LAB: Build staging and mart models on Redshift with incremental models, tests, snapshots and published docs.

dbt Overview 1

- Transformation layer in ELT
- dbt Core vs dbt Cloud
- Project structure: dbt_project.yml
- Profiles and connections

Materializations 2

- View
- Table
- Incremental
- Ephemeral

Reuse 3

- Macros and Jinja
- dbt packages
- Seeds
- Exposures

Models 4

- SQL models
- Python models
- Staging and marts layers
- Sources and ref
- The dbt DAG

Quality & Docs 5

- Generic tests
- Singular tests
- Documentation
- Lineage graphs

History & Deployment 6

- Snapshots for SCD Type 2
- Jobs and environments
- dbt on Snowflake
- dbt on Redshift

6

PHASE 6 OF 7 | MODULES 23-26 | 11 SESSIONS

Governance, Monitoring & DataOps

Lake Formation, SageMaker Catalog, CloudWatch, CloudTrail, IaC and CI/CD

23

MODULE 23 ·

Data Governance: Lake Formation & SageMaker Catalog

3 SESSIONS
16 TOPICS

DEA-C01
ALIGNED

Fine-grained access, business catalogs and PII discovery across the AWS data estate. LAB: Grant column- and row-filtered access with LF-tags, publish a dataset to SageMaker Catalog and scan the lake with Macie.

AWS Lake Formation 1

- Registering data locations
- Grants on databases, tables, columns
- LF-tags (tag-based access control)
- Data filters: row and cell level
- Cross-account sharing
- Permissions for Athena, Redshift, EMR

SageMaker Unified Studio 2

- Domains and domain units
- Projects
- SageMaker Catalog: business data catalog
- Data access through catalog projects
- Data lineage

Privacy & Compliance 3

- Identifying PII with Macie
- Masking and anonymisation
- AWS Config for configuration changes
- AWS Data Exchange
- Governance frameworks and sharing patterns

24

MODULE 24 · GREENS CORE

CloudWatch

2 SESSIONS
20 TOPICS

DEA-C01
ALIGNED

Monitoring, logging and alerting for every pipeline component. LAB: Build a pipeline dashboard with Logs Insights, metric filters, composite alarms and SNS alerts.

Overview 1

- Metrics
- Logs, log groups, log streams
- Custom metrics
- Dashboards

Alarms 2

- Alarms
- Composite alarms
- Anomaly detection
- Alerting and notifications

Logs 3

- Logs Insights queries
- Metric filters
- Subscription filters
- Retention policies

MODULE 25 · GREENS CORE

25 CloudTrail

2 SESSIONS
15 TOPICS

DEA-C01 ALIGNED

Auditing every API call for security, troubleshooting and compliance. LAB: Create an organization trail, query it with Athena and CloudTrail Lake, and alert on S3 bucket-policy changes.

CloudTrail Basics 1

- CloudTrail and API activity
- Management events vs data events
- Event history and trails
- Multi-Region and organization trails

Logging 2

- S3 activity logging
- IAM activity logging
- Log file integrity validation
- Delivery to S3 and CloudWatch Logs

Analysis 3

- CloudTrail Lake SQL queries
- Insights events
- Querying trails with Athena
- Security auditing and troubleshooting

Related Services 4

- EventBridge alerts on API calls
- AWS Config
- VPC Flow Logs

MODULE 26 ·

26 DataOps: Infrastructure as Code & CI/ CD

4 SESSIONS
17 TOPICS

DEA-C01 ALIGNED

Shipping data platforms as code with repeatable, tested deployments. LAB: Deploy S3, Glue jobs and a Step Functions workflow with CDK through a CodePipeline (or GitHub Actions) pipeline.

IaC on AWS 1

- AWS CloudFormation
- AWS CDK (Python)
- AWS SAM for serverless pipelines
- Terraform on AWS
- Repeatable multi-environment deploys

CI/CD 2

- Git branching and pull requests
- CodePipeline and CodeBuild
- CodeDeploy overview
- GitHub Actions with OIDC to AWS
- Testing data pipelines in CI

Containers 3

- Docker images for jobs
- Amazon ECR
- ECS and Fargate
- EKS overview
- AWS Batch

Productivity 4

- Amazon Q Developer for code
- AWS CLI and CloudShell

7

PHASE 7 OF 7 | MODULES 27-29 | 10 SESSIONS

BI & GenAI

Power BI, Amazon Quick Sight and GenAI with Bedrock and vectors

27

MODULE 27 · GREENS CORE

Power BI

5 SESSIONS
27 TOPICS

PL-300 ALIGNED

Turning AWS and Snowflake data into governed, fast reports. LAB: Build a sales report on Redshift with a star model, DAX time intelligence, RLS and scheduled refresh.

Overview 1

- Desktop, Service and Mobile
- PL-300 certification alignment

Data Modelling 2

- Star schema
- Relationships
- Cardinality

Security & Sharing 3

- Row-level security (RLS)
- Workspaces
- Publishing and sharing
- Deployment pipelines

Power Query 4

- Connecting to sources
- Data cleaning with M

DAX 5

- Calculated columns
- Measures
- Time intelligence

Refresh 6

- Scheduled refresh
- On-premises data gateways

AWS Sources 7

- Snowflake
- Redshift
- Athena
- S3 data lakes

Reports 8

- Visuals and report design
- Dashboards
- Apps

Performance 9

- Query folding
- Aggregations
- Incremental refresh
- Performance Analyzer

MODULE 28 ·

28 Amazon Quick Sight

2 SESSIONS
8 TOPICS

DEA-C01 ALIGNED

AWS-native BI (formerly QuickSight), the visualisation service in the DEA-C01 scope. LAB: Publish an Athena-backed Quick Sight dashboard with SPICE refresh and row-level security.

Quick Sight Basics 1

- Amazon Quick and Quick Sight
- Datasets from Athena and Redshift
- SPICE vs direct query

Analysis 2

- Visuals and dashboards
- Calculated fields
- Data verification and cleaning

Security 3

- Row-level security
- Sharing and embedding

MODULE 29 ·

29 GenAI & Vectors for Data Engineers

3 SESSIONS
17 TOPICS

DEA-C01 ALIGNED

Preparing data for AI and using LLMs inside pipelines, new in the DEA-C01 v1.1 outline. LAB: Chunk and embed policy PDFs from S3 into a Bedrock knowledge base and call it from a Lambda-based pipeline step.

Foundations 1

- LLMs, tokens, embeddings
- Amazon Bedrock models
- Prompting for data tasks
- Responsible AI

Vectors 2

- Vectorisation concepts
- Vector index types: HNSW vs IVF
- pgvector on Aurora
- OpenSearch vector search
- S3 Vectors overview

Bedrock Knowledge Bases 3

- Chunking strategies
- Sync from S3
- RAG retrieval
- Metadata filtering

CAPSTONE PROJECTS

Four projects. Four industries. One portfolio.

Each capstone is built end to end in your own AWS account, published on GitHub with an architecture diagram and README, and presented in a live demo with trainer review.

CAPSTONE01 · RETAIL & E-COMMERCE

Retail Lakehouse on S3, Glue & Iceberg

A governed lakehouse that unifies orders, customers, products and inventory from multiple systems.

S3 Glue Glue Crawler Apache Iceberg Lake Formation Athena Quick Sight

1. DMS and API ingestion into the raw layer
2. Incremental Glue jobs with bookmarks and DQDL
3. Iceberg silver tables with MERGE and SCD2
4. Gold KPIs partitioned for Athena
5. LF-tag security on PII columns
6. Quick Sight dashboard

CAPSTONE02 · BANKING & PAYMENTS

Real-Time Clickstream & Fraud Alerts

Events are processed in seconds, with suspicious patterns pushed to the operations team.

Kinesis Data Streams Managed Flink Firehose Lambda DynamoDB SNS CloudWatch

1. Python producer streaming card events
2. Flink windows for velocity and geo rules
3. Firehose delivery to S3 in Parquet
4. Lambda enrichment with DynamoDB lookups
5. SNS alerts and CloudWatch dashboard
6. Replay test from stream retention

CAPSTONE03 · HEALTHCARE & INSURANCE

Enterprise Warehouse with Redshift, dbt & Airflow

A claims warehouse loaded by CDC, modelled with dbt and orchestrated on MWAA.

Aurora AWS DMS Redshift dbt Amazon MWAA Power BI

1. CDC from Aurora with DMS to S3
2. COPY into Redshift staging
3. dbt staging, marts and SCD2 snapshots
4. Airflow 3 DAG with asset scheduling
5. RLS and masking for PHI
6. Power BI report with RLS

CAPSTONE04 · CROSS-INDUSTRY DATAOPS

Platform-as-Code with a GenAI Data Assistant

The whole platform built from code, deployed by CI/CD, audited and AI-enabled.

AWS CDK / Terraform CodePipeline / GitHub Actions Step Functions CloudTrail Bedrock OpenSearch

1. CDK stacks for lake, Glue and workflows per
2. CI pipeline with tests and approvals
3. Step Functions orchestration with retries
4. CloudTrail Lake audit queries and alarms
5. Bedrock knowledge base over the data catalog
6. Architecture review and live demo

Build a practical portfolio

The program's four end-to-end capstone builds across retail, banking, healthcare and DataOps.

01

Retail lakehouse on AWS

Ingest orders and customers into S3, catalogue with Glue, store as Iceberg tables and query with Athena.

02

Real-time clickstream and fraud alerts

Stream banking events with Kinesis, process with Managed Flink and alert on suspicious patterns in seconds.

03

Enterprise warehouse with Redshift and dbt

Load healthcare claims by CDC, model with dbt, orchestrate with Airflow and serve marts from Redshift.

04

Platform-as-Code with GenAI

Deploy the whole platform with CDK, Terraform and CI/CD, then add a Bedrock-powered data assistant.

Your portfolio tells the story

Show the pipeline design, code, data quality checks and dashboard. Explain the business question and your technical choices.

One stack. Four certification paths.

Use the course topics as a starting point for further certification study. Each provider sets its own exam requirements.

01

AWS Certified Data Engineer - Associate (DEA-C01)

AWS services, data ingestion, transformation, storage and pipeline operations.

02

Snowflake SnowPro Core

Snowflake architecture, data loading, virtual warehouses and platform fundamentals.

03

Microsoft Power BI Data Analyst (PL-300)

Data preparation, data modelling, DAX and reporting in Power BI.

04

Databricks Certified Data Engineer Associate

Further study in Spark and lakehouse concepts, building on the PySpark and EMR modules.

Course completion certificate

Complete the course to earn a Greens Technologies certificate of completion.

Provider certification exams are optional. Exam fees are separate. The course does not award these provider credentials automatically.

A practical data engineering toolkit

Learn how each tool fits into one connected workflow, from source data to business reporting.

SQL

Query, join and model business data.

Python

Read files and APIs; build reusable data scripts.

AWS

S3, Glue, EMR, Athena, Kinesis, Lambda and Redshift.

PySpark

Transform and process data with Apache Spark.

Snowflake

Load, store and query a cloud data warehouse.

dbt

Build, test and document SQL transformations.

Power BI

Create data models, reports and dashboards.

ALSO IN THE LABS

Linux, Git, GitHub, Airflow on MWAA, Step Functions, Lake Formation, Terraform, QuickSight and Bedrock.

The end-to-end flow

Source data > S3 > PySpark > Snowflake > dbt > Power BI

Where data skills can take you

Build toward data engineering and analytics roles with a connected cloud data portfolio.

Career path

Junior Data Engineer > Data Engineer > Senior Data Engineer > Lead Data Engineer

Salary range

Rs 10L to 45+L

per year

Indicative salary range. Actual offers vary by role and experience.

Companies hiring data engineering talent

PRODUCT-BASED

SERVICE-BASED

Company names show examples of employers for these skills, not a list of placement partners or a promise of placement at a specific company.

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Build your skills and prepare for the next step with Greens Technologys.

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Present your tools, projects and skills clearly, with trainer review.

02

Mock interviews

Practise technical answers and project explanations, then use the feedback.

03

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04

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